Task = 1

Program.1 :-

```
# include <iostream>
Using namespace std;
int main() {
    char letter;
    letter = 'A';
Cout << letter << "\n";
letter = 'B';
Cout << letter << "\n";
Cin >> letter;
return 0;
}
```

```
4
5 int main()
6 {
7     char letter;
8     letter = 'A';
9     cout << letter << "\n";
10
11    letter = 'B';
12    cout << letter << "\n";
13
14    cin >> letter;
15
16    return 0;
17 }
18
19
20
```

Compile Result

A
B

Program = 2

```
#include<iostream>
using namespace std;
int main()
{
  char letter;
  letter = 65;
cout<<letter<<'\n';
  letter = 66;
cout<<letter<<'\n';
  return 0;
}
```

Coding C++

Auto saved at 16:25:30

RUN MENU

```cpp
#include <iostream>
using namespace std;
int main()
{
    char letter;

    letter = 65;
    cout << letter << '\n';

    letter = 66;
    cout << letter << '\n';

    return 0;
}
```

Compile Result

```
A
B

[Process completed - press Enter]
```

Program = 3

```
#include <iostream>
using namespace std;
int main()
{
  letter = 'A';
cout << letter << '\n';
  letter = 'B';
cout << letter << '\n';
cin >> letter;
  return 0;
```

Coding C++

Auto saved at 16:26:35

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int letter;

    letter = 'A';
    cout << letter << '\n';

    letter = 'B';
    cout << letter << '\n';

    cin >> letter;

    return 0;
}
```

Compile Result

```
65
66
```

:- Task :02 :-

```
# include <iostream>
using namespace std;
  int main () {
      int num1= 10;
      int num2= 5;
cout << "First number =10" << num1 << endl;
cout << "Second number=5" << num2 << endl;
cout << "Addition=" << num1 + num2 << endl;
cout << "subtraction=" << num1 - num2 << endl;
cout << "Multiplication=" << num1 x num2 << endl;
   if (num2 != 0)
cout << "Division=" << (double) num1/num2 << endl;
   else
cout << "Divison is not possible." endl;
   return 0;
}
```

```
std;

int main()
{

    int num1 = 10;
    int num2 = 5;

    cout << "First number = " << num1 << endl;
    cout << "Second number = " << num2 << endl;

    cout << "Addition = " << num1 + num2 << endl;
    cout << "Subtraction = " << num1 - num2 << endl;
    cout << "Multiplication = " << num1 * num2 << endl;

    if (num2 != 0)
        cout << "Division = " << (double)num1 / num2 << endl;
    else
        cout << "Division is not possible." << endl;

    return 0;
}
```

Compile Result

```
First number = 10
Second number = 5
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2

[Process completed - press Enter]
```

Task = 03

```
#include <iostream>
using namespace std;
int main() {
    int val1 = 10;
    int val2 = 20;
    int val3 = 30;
    cout << "First number " << val1 << endl;
cout << "Second number" << val2 << endl;
cout << "Third number" << val3 << endl;
cout << "Reverse order:" << endl;
cout << val3 << endl;
cout << val2 << endl;
cout << val1 << endl;
    return 0;
}
```

Coding C++

Auto saved at 15:00:15

RUN MENU

```cpp

#include <iostream>
using namespace std;

int main()
{
    int val1 = 10;
    int val2 = 20;
    int val3 = 30;

    cout << "First number = " << val1 << endl;
    cout << "Second number = " << val2 << endl;
    cout << "Third number = " << val3 << endl;

    cout << "Reverse order:" << endl;
    cout << val3 << endl;
    cout << val2 << endl;
    cout << val1 << endl;

    return 0;
}

```

Compile Result

```
First number = 10
Second number = 20
Third number = 30
Reverse order:
30
20
10

[Process completed - press Enter]
```

Task = 04

```
#include <iostream>
#include <cmath>
#include <iomanip>
using namespace std;
int main()
{
    double x1=1, y1=2;
    double x2=4, y2=6;
    double distance = sqrt(pow(x2-x1,2)+
    pow(y2-y1,2));
cout<<fixed<<setprecision(2);
Cout<<"Distance"<<distance<<endl;
    return 0;
}
```

```cpp
#include <iostream>
#include <cmath>
#include <iomanip>
using namespace std;

int main()
{
    double x1 = 1, y1 = 2;
    double x2 = 4, y2 = 6;

    double distance = sqrt(pow(x2 - x1, 2) +
                           pow(y2 - y1, 2));

    cout << fixed << setprecision(2);
    cout << "Distance = " << distance << endl;

    return 0;
}
```

Compile Result

```
Distance = 5.00

[Process completed - press Enter]
```

Task = 5

(1) Area of Circle

```
#include <iostream>
#include <iomanip>
using namespace std;
int main()
{
  double radius=5;
  const double PI =3.14159;
  double area = PI * radius x radius;
  Cout fixed << setprecision(2);
  Cout << "Radius=" << radius << endl;
  Cout << "Area of circle=" << area << endl;
    return 0;
```

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    double radius = 5;
    const double PI = 3.14159;

    double area = PI * radius * radius;

    cout << fixed << setprecision(2);
    cout << "Radius = " << radius << endl;
    cout << "Area of Circle = " << area << endl;

    return 0;
}
```

Compile Result

```
Radius = 5.00
Area of Circle = 78.54

[Process completed - press Enter]
```

(e) Area of Rectangle

```
#include <iostream>
using namespace std;
int main(){
int length = 10;
int width = 5;
int area = length x width;
cout<< "length = " << length << endl;
cout<< "width = " << width << endl;
cout<< Area of Rectangle = " << area << endl;
return 0;
}
```

Coding C++

Auto saved at 17:06:04

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int length = 10;
    int width = 5;

    int area = length * width;

    cout << "Length = " << length << endl;
    cout << "Width = " << width << endl;
    cout << "Area of Rectangle = " << area << endl;

    return 0;
}
```

Compile Result

```
Length = 10
Width = 5
Area of Rectangle = 50

[Process completed - press Enter]
```

(3) Area of Triangle

```
# include <iostream>
  Using name space std;
int main()
{
   int base =10;
   int height = 6;
double area = 0.5 x base x height;
Cout<<"Base =" << base << endl;
Cout<<"Height= " << height<<endl;
Cout << "Area of Friangle=" << area<<endl;
   return 0;
  }
```

Coding C++

Auto saved at 17:07:10

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int base = 10;
    int height = 6;

    double area = 0.5 * base * height;

    cout << "Base = " << base << endl;
    cout << "Height = " << height << endl;
    cout << "Area of Triangle = " << area << endl;

    return 0;
}
```

Compile Result

```
Base = 10
Height = 6
Area of Triangle = 30

[Process completed - press Enter]
```

Bonus Task

```
#include<iostream>
using namespace std;
int main()
{
    int number = 12345;
Cout<<" original Number ="<<number<<endl
Cout<<number/10000<<" ";
Cout<<(number/1000)%10<<" ";
Cout<<(number/100)%10<<" ";
Cout<<(number/10)%10<<" ";
Cout<<number%10<<endl;
    return 0;
}
```

Coding C++

Auto saved at 17:08:18

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number = 12345;

    cout << "Original Number = " << number << endl;

    cout << number / 10000 << "    ";
    cout << (number / 1000) % 10 << "    ";
    cout << (number / 100) % 10 << "    ";
    cout << (number / 10) % 10 << "    ";
    cout << number % 10 << endl;

    return 0;
}
```

Compile Result

```
Original Number = 12345
1   2   3   4   5

[Process completed - press Enter]
```

Home Task

(1)

```
#include<iostream>
using namespace std;
int main()
{
    int var1 =10;
    int var2 =20;
    int temp ;
Cout << "Before swapping:" << endl;
Cout << "var1 = " << var1 << endl;
Cout << "var2 = " << var2 << endl;
        temp = var1
        var1 = var2
        var2 = temp;
Cout << "After swapping:" << endl;
Cout << " var1 = " << var1 << endl;
Cout << "var2 = " << var2 << endl;
    return 0;
}
```

Coding C++

Auto saved at 17:09:12

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1 = 10;
    int var2 = 20;
    int temp;

    cout << "Before Swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After Swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

Compile Result

```
Before Swapping:
var1 = 10
var2 = 20
After Swapping:
var1 = 20
var2 = 10

[Process completed - press Enter]
```

(2)

```
#include<iostream>
using namespace std;
int main()
{
    char letter='b';
cout<<"Lowercase letter="<<letter<<endl;
    letter=letter-32;
cout<<"Uppercase letter="<<letter<<endl;
    return 0;
}
```

Coding C++

Auto saved at 17:10:26

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter = 'b';

    cout << "Lowercase Letter = " << letter << endl;

    letter = letter - 32;

    cout << "Uppercase Letter = " << letter << endl;

    return 0;
}
```

Compile Result

```
Lowercase Letter = b
Uppercase Letter = B

[Process completed - press Enter]
```